SOFTWARE ENGINEERING LABORATORY · EXPERIMENT 3

Data Flow Diagrams (Context, Level-1 & Level-2)

Project: Orbital Sentinel — Satellite Telemetry Anomaly Detection with Live Orbital Tracking

Modelled in draw.io, consistent with the approved SRS (Experiment 1)

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1. Aim

To analyse the functional requirements of the Orbital Sentinel system and develop its Context Diagram (Level-0 DFD) together with Level-1 and Level-2 Data Flow Diagrams, based on the Software Requirements Specification prepared in the earlier experiments. The diagrams were created in draw.io using standard DFD notation — external entities as rectangles, processes as circles, and data stores as open-ended rectangles.

2. Project Title

Orbital Sentinel — an early-warning health-monitoring system for satellite fleets. It learns the normal behaviour of a spacecraft's telemetry, flags abnormal behaviour as readings arrive, identifies the responsible sensor channel, ranks satellites by risk, and presents everything on a live 3D globe driven by public orbital data.

3. Functional Requirements

#	Functionality	Description
FR-1	Telemetry ingestion & replay	Load and stream spacecraft telemetry in time order.
FR-2	Anomaly detection	Flag abnormal behaviour using a baseline detector with dynamic thresholding.
FR-3	Anomaly explanation	Attribute each anomaly to the responsible sensor channel(s).
FR-4	Risk prioritisation	Suppress transient spikes, score severity, and rank satellites by risk.
FR-5	Live orbital tracking	Fetch orbital elements and compute positions via SGP4.
FR-6	Visualisation	Render a 3D globe with a health panel and ranked alert list.
FR-7	Alerting & dashboard	Present verdicts and a risk-ordered alert list to the analyst.
FR-8	Detection logging	Record every detection with its metadata for later review.
FR-9	Configuration & admin	Manage monitored satellites and tune detection settings.

4. External Entities

Entity	Role
Analyst	Monitors the globe and alerts and inspects flagged satellites.
Administrator	Configures monitored satellites and detection settings.
Evaluator	Reviews detection logs and accuracy metrics.
Telemetry Dataset	External data source providing raw spacecraft telemetry (SMAP/MSL).

TLE Feed	External service providing orbital element sets (CelesTrak).

5. Context Diagram (Level-0 DFD)

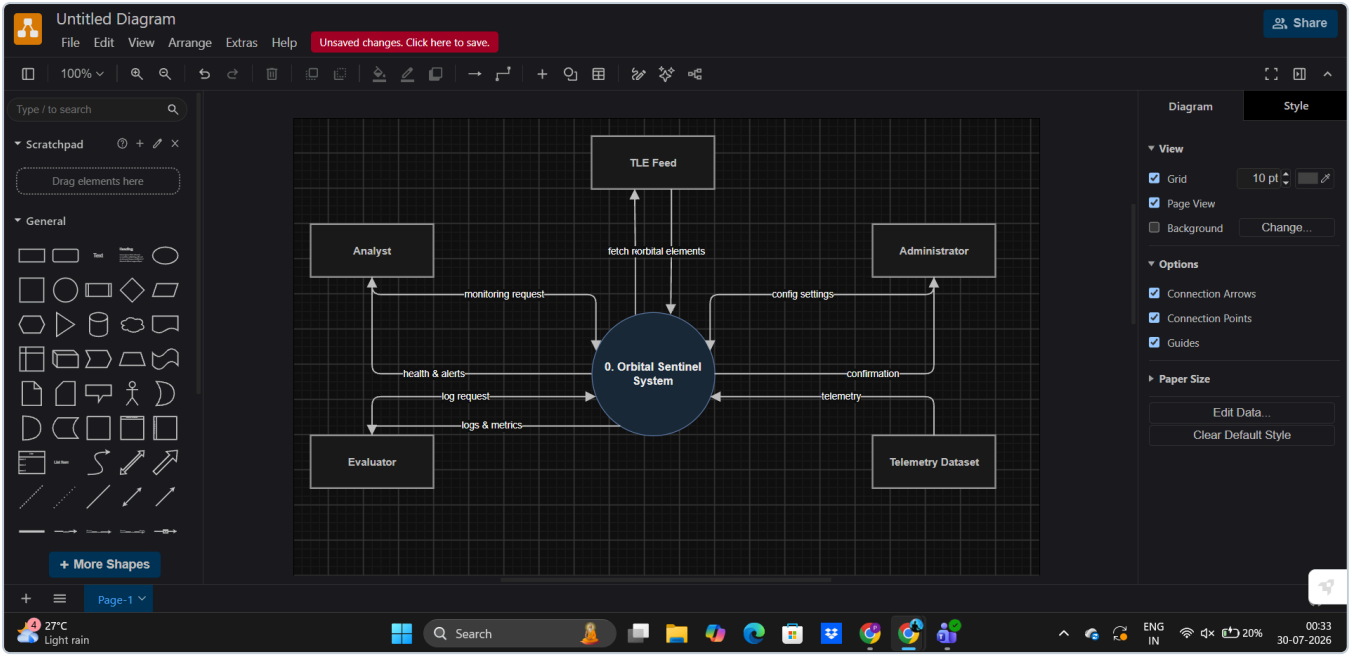


Figure 5.1 — Context Diagram of Orbital Sentinel (draw.io, timestamped 30-07-2026).

Explanation. The context diagram represents the entire system as a single process that exchanges data with its three human users and two external data sources. Telemetry and orbital elements flow in; health verdicts, alerts, and logs flow out. As required at Level-0, there is exactly one process and no data stores.

6. Level-1 DFD

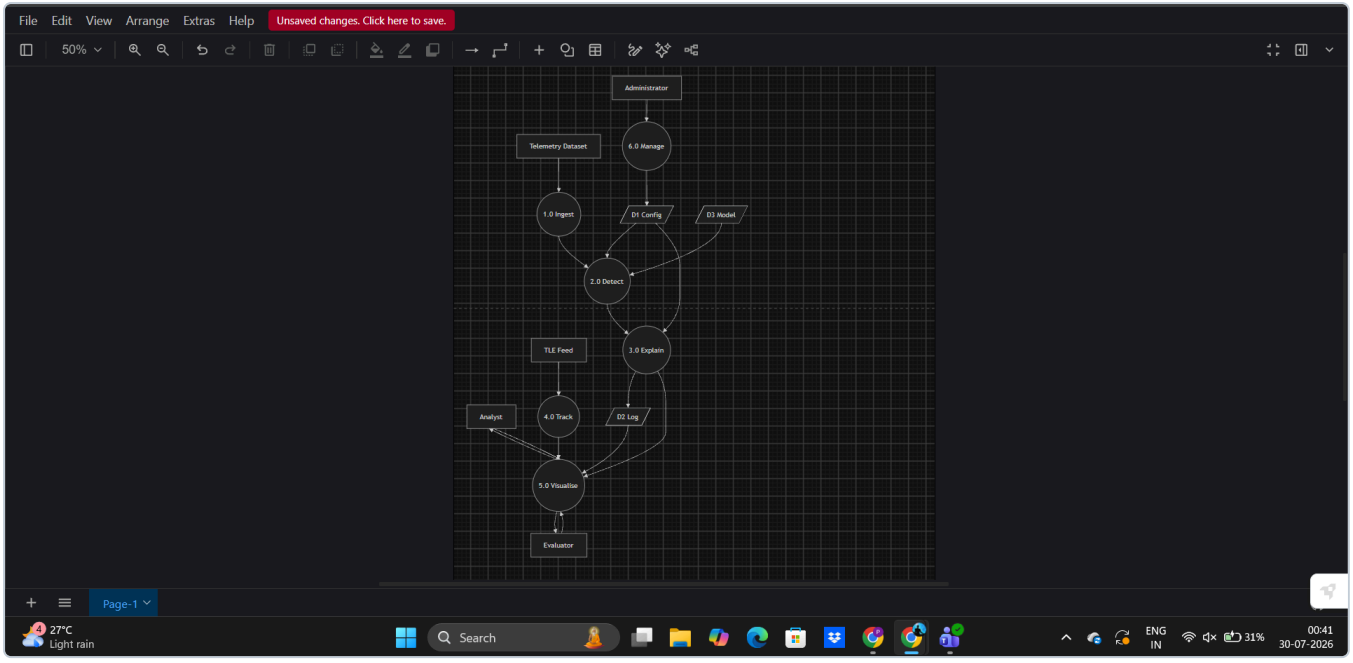


Figure 6.1 — Level-1 DFD of Orbital Sentinel (draw.io, timestamped 30-07-2026).

Explanation. The single process is decomposed into six major processes (1.0 Ingest, 2.0 Detect, 3.0 Explain, 4.0 Track, 5.0 Visualise, 6.0 Manage) and three data stores (D1 Config, D2 Log, D3 Model). It shows the two parallel paths — the telemetry/detection path and the orbital-tracking path — converging at the dashboard, with configuration read from and written to the stores.

7. Level-2 DFD (Process 3.0 — Explain & Prioritise)

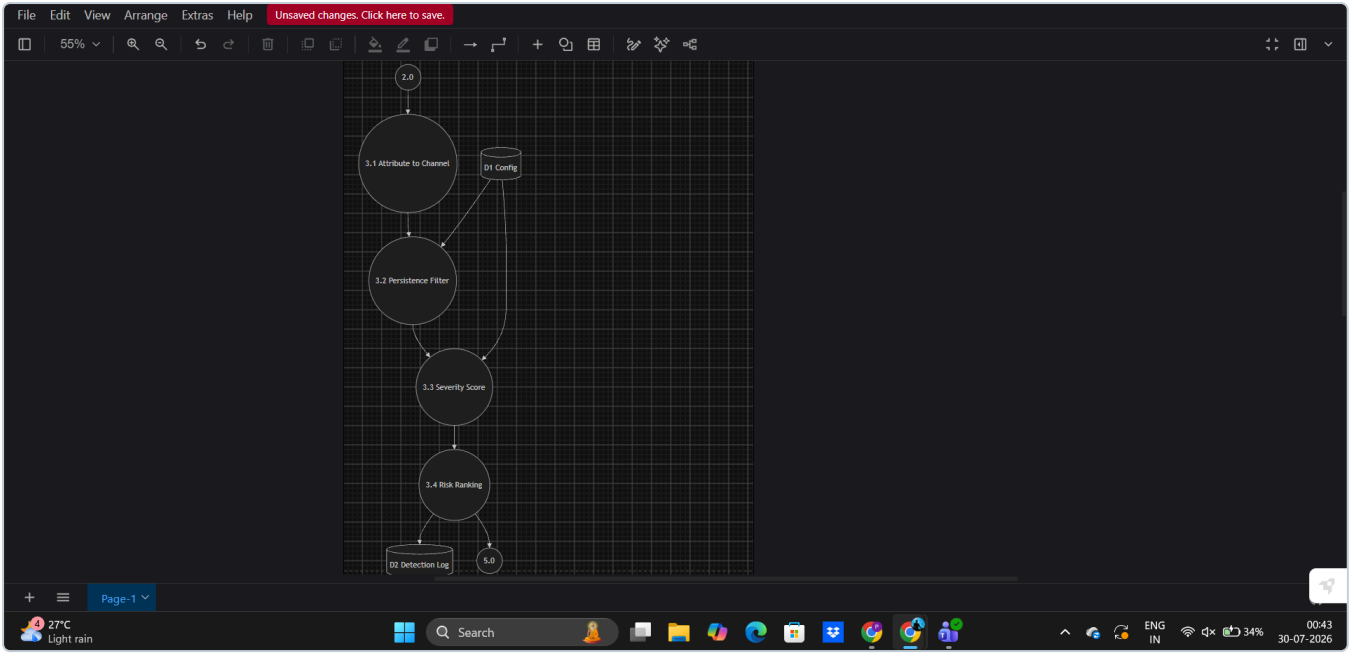


Figure 7.1 — Level-2 DFD expanding process 3.0 (draw.io, timestamped 30-07-2026).

Explanation. Process 3.0 is expanded into four subprocesses. A detected anomaly is first attributed to the responsible channel (3.1), then filtered to remove one-off transient spikes (3.2), scored for severity (3.3), and finally used to update the satellite's risk ranking (3.4), which is logged to D2 and passed on to the dashboard (process 5.0). This is the project's novelty layer shown in detail.

8. DFD Validation

The diagrams were checked against the standard DFD validation rules:

	Rule	Status
✓	Every process has at least one input and one output	Satisfied — no process is a source or sink only.
✓	All data flows are named	Satisfied — each arrow carries a label.
✓	No direct flow between two external entities	Satisfied — entities communicate only via the system.
✓	No direct flow between two data stores	Satisfied — stores connect only through processes.
✓	Parent–child balancing (Context ↔ L1 ↔ L2)	Satisfied — see below.

Balancing note. The inputs and outputs of process 3.0 in the Level-1 diagram (anomalies in from 2.0; verdict and risk ranking out to 5.0; a log written to D2; and configuration read from D1) reappear exactly at the boundary of the Level-2 diagram, so the decomposition is balanced. The same holds between the Context diagram and Level-1: the external flows to and from the five entities are preserved.

9. Result and Inference

The functional requirements and external entities of Orbital Sentinel were identified from the SRS, and the Context, Level-1, and Level-2 Data Flow Diagrams were successfully constructed in draw.io using standard notation and verified against the validation rules. The exercise confirmed that the system decomposes cleanly into a detection path and an orbital-tracking path that meet at the dashboard, and that the novelty process (explanation and prioritisation) expands naturally into four well-defined subprocesses while preserving parent–child balancing across all three levels.